

Debodeep Banerjee

Research Profile

PhD researcher in Artificial Intelligence at the University of Pisa and the University of Trento. My research focuses on hybrid human-machine learning, decision-making under uncertainty, and the integration of vision-language models into human-centered AI systems. I am particularly interested in designing learning frameworks that incorporate human feedback, interpretability, and statistical guarantees.

Research Interests

Human-in-the-loop learning
Hybrid decision-making systems
Natural language processing
Large language models
Vision-language models

Education

2022–Present **PhD in Artificial Intelligence**, *University of Pisa & University of Trento*
2022 **Master in Data Science**, *Sapienza University of Rome*
2018 **M.Sc. in Statistics**, *University of Madras*
2016 **B.Sc. in Statistics (Hons)**, *Asutosh College, University of Calcutta*

Work Experience

Jun–Oct 2022 **Data Science Intern**, *Project Consulting SRL*, Trento
Jul 2021–Jan 2022 **AI Intern**, *Baker Hughes*, Italy
May–Aug 2018 **Subject Matter Expert**, *SPI Global*, India
May–Jun 2017 **Research Intern**, *Indian Statistical Institute*, India

Publications

Conference and Workshop Papers

- 2025 Banerjee, D. et al. “MedGellan: LLM-Generated Medical Guidance to Support Physicians.”, HLDM Workshop at ECML-PKDD 2025
Under Review / Submitted
2025 Banerjee, D. et al. “Learning to Guide Human Decision Makers with Vision–Language Models.”, Transactions on Machine Learning Research (Under Review)
2025 Banerjee, D. et al. “Hybrid Decision Making via Conformal VLM-generated Guidance.”, HHA1 2025 (Submitted)

Presentations

- 2018 International Conference on Theory and Application of Statistics and Information Science – Bayesian analysis of $1 \times j$ contingency table
- 2018 National Conference at Marian College – CSR and Sustainable Development in India

Research Projects

- Master in Data Science **Title:** Nonparametric Modal Regression for Directional Responses
Supervisor: Prof. Pierpaolo Brutti
Developed EM-based modal regression algorithm for directional predictors and linear responses.
- M.Sc. Statistics **Title:** A Study on Classical Inference on Single Binomial Proportion
Supervisor: Dr. Subbiah M.
Focused on evaluating confidence intervals using coverage probability and expected run length.

Technical Report

- 2017–2018 **Title:** Comparative Analysis of Two $1 \times j$ Contingency Table under Bayesian Perspective
Authors: Debodeep Banerjee, Trina Sahoo, Dr. Subbiah M., Dr. M.R. Srinivasan
Published in University of Madras Library, used R and Monte Carlo simulation techniques.

Technical Expertise

- Programming Python, R, SQL
- Machine Learning PyTorch, scikit-learn, NLP, Vision–Language Models
- Gen-AI RAG, Ollama, Langgraph, vLLM
- Frameworks
- Statistics Inference, conformal prediction, regression modelling
- Tools LaTeX, Git

Achievements

- 1 First place in Inter-Departmental Quiz, University of Madras
- 2 First place in Inter-Collegiate Quiz, Madras Christian College

Languages

- Fluent Bengali, Hindi, English
- Basic Italian